

Methodological Adjustment: Analytical State Inference

In standard discrete-time Partially Observable Markov Decision Processes (POMDPs) formulated under the Free Energy Principle, a distinction is maintained between the inference of transient hidden states (s_t) and the learning of enduring parameters ($\mathbf{A}$, $\mathbf{B}$).

The original agent implementation parameterised the approximate posterior over hidden states as a Dirichlet distribution, $Q(s) = \text{Dir}(\theta)$, and minimised Variational Free Energy (VFE) at each timestep via stochastic gradient descent (SGD) using the Adam optimiser over sampled expectations. We have adjusted this to compute the exact analytical posterior using standard Categorical message passing.

This adjustment is justified on three technical grounds:

1. VFE Minimisation and Tractability Variational inference is typically employed when the true posterior, $P(s|o)$, is analytically intractable. The objective is to minimise the VFE functional, which is defined as the Kullback-Leibler (KL) divergence between the approximate and true posterior, minus the log marginal likelihood: $F = D_{KL}[Q(s) \parallel P(s|o)] - \ln P(o)$. Because the iterated normal-form game relies on a finite, low-dimensional discrete state space, the true posterior is entirely tractable. By calculating the exact posterior analytically, we set $Q(s) = P(s|o)$, driving the KL divergence term strictly to zero. Therefore, exact Bayesian inference does not abandon VFE minimisation; rather, it achieves the exact global minimum of the VFE functional in a single computational pass, eliminating the noise and local minima inherent to SGD.

2. Ontological Consistency of Uncertainty In the canonical literature on discrete Active Inference (e.g., Friston et al., 2017; Da Costa et al., 2020), there is a strict mapping between the temporal lifespan of a variable and its probability distribution. First-order uncertainty (Categorical distributions) is applied to transient hidden states, as the agent receives only a single observation before the state transitions. Second-order uncertainty (Dirichlet distributions) is reserved for enduring variables, such as the transition and observation matrices, allowing the agent to accumulate evidence and concentrate its parameters over multiple timesteps. Applying a Dirichlet distribution to a transient state is statistically underdetermined, as the agent cannot accumulate multiple observations to refine a second-order belief about a momentary state.

3. Computational Efficiency The exact update— $s_t = \sigma(\ln \mathbf{A}^T o_t + \ln \mathbf{B} s_{t-1})$ —requires only basic matrix multiplication and a softmax operation. By removing the need for iterative sampling and gradient calculations at every perception step, the computational overhead is substantially reduced.

We note that the original use of differentiable Dirichlet sampling might have been intended as an architectural bridge to continuous state spaces or hierarchical models where states act as parameters for lower levels. However, within the boundaries of a purely discrete iterated game,

the analytical Categorical update represents the canonical, exact, and most efficient formulation of Active Inference perception.